Sentiment Analysis for Earning Calls

* NLP Final Project Report

Raul Silva Noreña
DSAI, EDHEC
raul.silvanorena@edhec.com

I. Introduction

This report presents an analysis of earnings call sentiment and its ability to predict subsequent stock market reactions, measured by cumulative abnormal returns from the Carhart 4 factor model. The objective is to build multiple sentiment measures from earnings call transcripts using both traditional Bag-of-Words (BoW) approaches and modern large language models (LLMs), compare how these measures relate to one another, and evaluate which one best captures the informational content that moves stock prices after an earnings call.

A. Problem Definition

To address this question, we adopt an experimental approach, testing eight different sentiment extraction methods across three distinct text units: the management presentation, the management Q&A answers, and the full call transcript. On the dictionary side, we compare general-purpose and finance-specific lexicons, with and without negation handling, as well as a custom dictionary built from the corpus itself using word embeddings. On the LLM side, we deploy a pre-trained financial sentiment model (FinBERT), a distilled version of it, and a model fine-tuned directly on stock returns. This produces a total of 24 sentiment measures, which we then evaluate through correlation analysis, univariate regressions, and controlled regressions that account for hard financial information such as earnings surprises, analyst coverage, and historical volatility. The following section details the methodology behind each sentiment measure.

To assess the predictive value of each measure, we rely on ordinary least squares (OLS) regressions of abnormal returns on sentiment scores. We first examine univariate relationships to identify which measures carry a statistically significant association with market reactions, and then introduce control variables including; earnings surprises, analyst coverage, and historical volatility, in order to test whether sentiment provides incremental explanatory power beyond simple financial information. The full specification of these regressions, along with the correlation analysis between measures, is detailed in the methodology section that follows.

II. Sentiment Measures

A. Bag-of-words Methods

The first sentiment measure uses the Harvard General Inquirer (GI) dictionary used in class, a general-purpose lexicon that classifies words as positive or negative. For each transcript, the score is computed as the difference between positive and negative word counts, normalized by total word count. While widely used in text analysis, this dictionary was not designed for financial language, meaning words such as ”liability” or ”capital” may be misclassified as negative when they carry neutral meaning in an earnings call context. To address one known limitation of simple dictionary lookups, we also compute a negation-aware variant of the Harvard GI score. This version applies a three-word lookback window: if a sentiment word is preceded by a negation term (e.g., ”not”, ”never”, ”without”), its polarity is flipped, so that phrases like ”not strong” are correctly counted as negative rather than positive.

The third and fourth measures replace the Harvard GI with the Loughran-McDonald (LM) dictionary, which was specifically developed for financial text, using the most recent version, compared to the version used throughout the course. It correctly treats this domain-specific terms like ”liability”, ”tax”, or ”cost” as neutral rather than negative, making it a more appropriate lexicon for earnings call analysis. As with Harvard GI, we compute both a plain version and a negation-aware variant that flips polarity when sentiment words fall within a negation window.

The fifth measure takes a different approach by constructing a custom dictionary directly from the corpus. A Word2Vec model is trained on the earnings call transcripts, and a set of hand-picked finance seed words (e.g., ”growth”, ”beat”, ”decline”, ”miss”) is expanded by retrieving their nearest neighbours in the embedding space, retaining only those with cosine similarity above 0.65. Words appearing in both the positive and negative expanded lists are removed as ambiguous. This produces

a corpus-specific lexicon that reflects the actual vocabulary used in these calls, rather than relying on a pre-built general or financial word list.

B. Large Language Model Methods

Starting with the LLM measures, the sixth score uses FinBERT, a BERT model pre-trained on general text and fine-tuned on financial news for three-class sentiment classification (positive, negative or neutral). Since the earnings call transcripts exceed BERT's 512-token input limit, each transcript is split into 450-word chunks with a small overlap, and FinBERT produces a probability distribution over the three classes for each chunk. The final sentiment score is the average of P(positive) minus P(negative) across all chunks. Unlike the dictionary methods, FinBERT captures word order, context, and semantic nuance.

The seventh measure explores knowledge distillation: a smaller DistilBERT model is trained to replicate FinBERT's output. FinBERT first generates soft probability vectors for each transcript (e.g., [0.7, 0.1, 0.2]), and the student model learns to match these distributions by minimizing KL-divergence loss. The goal is to test whether a lighter and faster model can approximate the teacher's sentiment judgments with minimal information loss, which would be relevant for large-scale or real-time applications.

The eighth and final measure takes the most data-driven approach: a DistilBERT model is fine-tuned as a regressor to predict the market reaction (CAR-11-Carhart) directly from transcript text. A temporal train-test split is used (earliest 80% for training, latest 20% for testing), and the model is scored only on the out-of-sample portion to avoid data leakage. Here, the model's predicted CAR itself serves as the sentiment measure, it learns whatever textual signal best predicts stock returns, without being explicitly told what "sentiment" means. While this is the most flexible approach, it is also the most susceptible to overfitting given the limited sample size.

III. Results

All sentiment measures carry a positive coefficient in the univariate regressions against CAR-11-Carhart, consistent with the idea that more positive language during an earnings call is associated with a more favorable subsequent stock market reaction. However, the magnitude and statistical significance of this relationship varies considerably across methods and text units.

To start with the method comparison. The fine-tuned DistilBERT regressor achieves the highest univariate R^2 (2.22%), followed closely by the Loughran-McDonald dictionary with negation handling (1.85% on the full call) and plain Loughran-McDonald (1.78%), as shown in Figure 5. FinBERT ranks next, with R^2 values between 1.5% and 1.7% depending on the text unit. The custom Word2Vec dictionary and Harvard GI fall further behind, while the distilled model performs worst among the LLM-based approaches. The coefficient plot in Figure 3 illustrates this hierarchy across all three text units, where red bars indicate statistical significance at the 5% level. An important caveat applies to the fine-tuned model: because it is scored only on the out-of-sample test set (N = 564), its results are not directly comparable to those of the other methods, which are estimated on the full sample (N = 2,817). Its apparent advantage may partly reflect the smaller, potentially less noisy subsample rather than a genuinely superior extraction of sentiment.

Continuing with the finance-specific vs. general-purpose dictionaries. The single clearest finding among the Bag-of-Words methods is that the Loughran-McDonald dictionary vastly outperforms the Harvard General Inquirer. Across all three text units, LM sentiment is statistically significant at the 1% level with standardized betas around 0.006–0.008 (Table II), whereas Harvard GI coefficients are smaller and often insignificant, particularly on the presentation and full call texts (see Figure 3, left and right panels). This confirms that a general-purpose lexicon misclassifies too many domain-neutral terms as sentiment-bearing, diluting the signal in financial text.

Touching on negation awareness, this addition to the dictionaries produces only a marginal improvement. The correlation between LM and LM with negation is 0.99 within each text unit (Figure **??**), and their regression coefficients are nearly identical (Table II). While the negation-aware variant consistently ranks a bit higher, the difference is economically negligible. This suggests that explicit negation (e.g., "not profitable") is relatively rare in the structured, carefully worded language of earnings calls, limiting the scope for negation handling to add value.

On the side of LLMs vs. dictionaries, FinBERT performs comparably to the Loughran-McDonald dictionary but does not clearly dominate it. On the full call text, FinBERT achieves an R^2 of 1.52% versus 1.78% for plain LM (Figure 5) which is slightly lower despite its ability to capture context and word order. This may reflect the fact that FinBERT was fine-tuned on financial news rather than earnings call transcripts, and that the chunking procedure required to handle long documents introduces noise, which is a detaill that can be improved in further experiments. The distilled model, intended as a lighter approximation of FinBERT, loses nearly all predictive signal: its coefficients are barely significant and its R^2 values are among the lowest observed (Figure 5, Table II). This indicates that the compression inherent in

knowledge distillation discarded the nuances that made FinBERT useful in the first place.

For the dictionary-based methods in the text unit comparison, the full call transcript tends to produce the strongest results, followed by the presentation and then the management Q&A answers alone. This pattern is visible in Figure 2, where the green bars (full call) generally dominate across the LM and FinBERT method groups. For the fine-tuned model, however, the presentation alone performs as well as the full call (R^2 = 2.22% for both), suggesting that the prepared remarks already contain most of the predictive signal and that the Q&A adds relatively little incremental value when processed by a model trained to predict returns. The Q&A section on its own is the weakest text unit across nearly all methods, possibly because analyst questions introduce noise that is unrelated to management sentiment.

On the incremental value beyond controls, we decided to introduce; earnings surprise (SurpDec), analyst coverage (NUMEST), and historical volatility (hvol) as controls. The baseline model explains 7.7% of the variance in abnormal returns, driven primarily by SurpDec (Table I). Adding sentiment to this specification yields incremental R^2 gains ranging from near zero (Harvard GI on the presentation, distilled model) to 0.83 percentage points (fine-tuned model on the presentation), as ranked in Figure 4. The top five measures; fine-tuned (presentation and full call), LM with negation (full call), plain LM (full call), and FinBERT (presentation), all produce statistically significant sentiment coefficients in the controlled regression (Table I), confirming that textual sentiment captures information about market reactions that is not already embedded in hard financial variables. Importantly, the correlations between sentiment measures and control variables are low (generally below 0.18, as shown in Figure 7), indicating that sentiment is not merely a proxy for earnings surprises or volatility but reflects a distinct dimension of the information conveyed during the call.

Putting the magnitude of the results into perspective, the $R2^2$ values reported throughout this analysis are small in absolute terms, with the best univariate specification explaining just over 2% of the cross-sectional variance in abnormal returns. The scatter plots for the top three measures (Figure 1) illustrate this visually: while the fitted line slopes upward, the dispersion around it is substantial. This is, however, consistent with the broader asset pricing literature, where individual predictors rarely explain more than a few percent of return variation. The fact that a text-based measure achieves statistical significance alongside established financial controls is itself a meaningful result, suggesting that the qualitative tone of managerial communication carries real, if modest, informational content for equity markets, so adding the ”sentiment” factor in an well known factor model would be the natural next step for assessing the significance of this factor.

IV. Conclusion

This study compared eight sentiment extraction methods, ranging from simple dictionary lookups to fine-tuned transformer models, applied to three sections of earnings call transcripts, producing 24 distinct sentiment measures evaluated against post-call abnormal stock returns. The results point to a clear hierarchy: the Loughran-McDonald dictionary, with or without negation handling, consistently delivers the strongest and most reliable signal among the unsupervised methods, confirming that domain-specific lexicons are essential when working with financial text. FinBERT performs comparably but does not offer a decisive advantage over the dictionary approach, while the knowledge distillation experiment demonstrates that compressing a transformer into a smaller model can destroy the very nuances that make it useful. The fine-tuned regressor achieves the highest R^2 in both univariate and controlled specifications, but its smaller out-of-sample evaluation set warrants caution in drawing strong conclusions from this result.

Beyond the method comparison, two broader findings emerge. First, the management presentation and the full call transcript are more predictive than the Q&A section alone, suggesting that the prepared remarks already encode the most market-relevant sentiment. Second, and perhaps most importantly, all top-performing sentiment measures remain statistically significant after controlling for earnings surprises, analyst coverage, and historical volatility, with incremental R^2 gains of up to 0.83 percentage points. While modest in absolute terms, this confirms that the qualitative tone of managerial communication captures a dimension of information that hard financial variables do not, a finding consistent with the growing body of literature on the role of textual information in financial markets.

Appendix A
Additional Figures

Table I
Controlled Regressions: CAR-11-Carhart on Sentiment and Controls (Top-5 Measures)

	M1 (controls)	**Finetuned Presentation**	**Finetuned Full Call**	**LM_neg Full Call**	**LM Full Call**	**FinBERT Presentation**
surp	-0.0035	-0.0048	-0.0048	-0.0034	-0.0034	-0.0031
	(0.0018)	(0.0033)	(0.0033)	(0.0019)	(0.0019)	(0.0018)
SurpDec	$+0.0174^{***}$	$+0.0167^{***}$	$+0.0167^{***}$	$+0.0165^{***}$	$+0.0165^{***}$	$+0.0167^{***}$
	(0.0016)	(0.0037)	(0.0037)	(0.0016)	(0.0016)	(0.0016)
NUMEST	$+0.0010$	$+0.0034$	$+0.0034$	$+0.0002$	$+0.0002$	$+0.0006$
	(0.0012)	(0.0024)	(0.0024)	(0.0012)	(0.0012)	(0.0012)
hvol	-0.0058^{*}	-0.0114^{*}	-0.0114^{*}	-0.0052^{*}	-0.0053^{*}	-0.0050^{*}
	(0.0023)	(0.0050)	(0.0050)	(0.0024)	(0.0024)	(0.0024)
Sentiment		$+0.0056^{*}$	$+0.0056^{*}$	$+0.0050^{***}$	$+0.0049^{***}$	$+0.0046^{***}$
		(0.0027)	(0.0027)	(0.0012)	(0.0012)	(0.0011)
Constant	-0.0005	-0.0028	-0.0028	-0.0005	-0.0005	-0.0005
	(0.0011)	(0.0024)	(0.0024)	(0.0011)	(0.0011)	(0.0011)
N	2,773	554	554	2,773	2,773	2,773
R^2	0.0771					
R^2 (M1)		0.1096	0.1096	0.0771	0.0771	0.0771
R^2 (M3)		0.1179	0.1178	0.0835	0.0831	0.0826
ΔR^2		0.00829	0.00824	0.00638	0.00599	0.00552

HC3 heteroskedasticity-robust standard errors in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
The finetuned model is scored only on the out-of-sample test set ($N = 554$); its M1 baseline is estimated on the same subsample.

Table II
Univariate OLS Results: CAR-11-Carhart $\sim$ Sentiment (All 24 Measures)

Method	**Text Unit**	$\hat{\beta}$	t**-stat**	p**-value**	R^2	N
Fine-tuned DistilBERT (OOS only)						
Finetuned	Presentation	$+0.0089^{***}$	3.42	0.0006	0.0222	564
Finetuned	Full Call	$+0.0089^{***}$	3.41	0.0006	0.0222	564
Finetuned	Mgmt QA	$+0.0023$	0.96	0.3351	0.0015	564
Loughran-McDonald with negation						
LM_neg	Full Call	$+0.0083^{***}$	7.20	< 0.001	0.0185	2,817
LM_neg	Presentation	$+0.0071^{***}$	6.29	< 0.001	0.0137	2,817
LM_neg	Mgmt QA	$+0.0065^{***}$	5.75	< 0.001	0.0115	2,817
Loughran-McDonald (plain)						
LM	Full Call	$+0.0081^{***}$	7.09	< 0.001	0.0178	2,817
LM	Presentation	$+0.0070^{***}$	6.20	< 0.001	0.0132	2,817
LM	Mgmt QA	$+0.0064^{***}$	5.70	< 0.001	0.0111	2,817
FinBERT						
FinBERT	Full Call	$+0.0075^{***}$	6.89	< 0.001	0.0152	2,817
FinBERT	Presentation	$+0.0074^{***}$	6.60	< 0.001	0.0147	2,817
FinBERT	Mgmt QA	$+0.0046^{***}$	4.05	< 0.001	0.0056	2,817
Custom Word2Vec dictionary						
Custom	Full Call	$+0.0046^{***}$	4.14	< 0.001	0.0056	2,817
Custom	Presentation	$+0.0045^{***}$	4.02	< 0.001	0.0053	2,817
Custom	Mgmt QA	$+0.0031^{**}$	2.83	0.0047	0.0025	2,817
Harvard General Inquirer						
HV	Mgmt QA	$+0.0036^{**}$	3.13	0.0018	0.0035	2,817
HV	Full Call	$+0.0022$	1.90	0.0581	0.0013	2,817
HV	Presentation	$+0.0013$	1.08	0.2797	0.0004	2,817
Harvard General Inquirer with negation						
HV_neg	Mgmt QA	$+0.0035^{**}$	3.07	0.0022	0.0033	2,817
HV_neg	Full Call	$+0.0022$	1.91	0.0556	0.0013	2,817
HV_neg	Presentation	$+0.0013$	1.09	0.2770	0.0004	2,817
Distilled DistilBERT (Knowledge Distillation)						
Distilled	Full Call	$+0.0025^{*}$	2.19	0.0288	0.0017	2,817
Distilled	Presentation	$+0.0023^{*}$	2.01	0.0442	0.0015	2,817
Distilled	Mgmt QA	$+0.0003$	0.28	0.7819	0.0000	2,817

Dependent variable: CAR-11-Carhart. Sentiment scores are standardised (z-score) before estimation. HC3 heteroskedasticity-robust standard errors used throughout.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
The finetuned model is evaluated on the out-of-sample test set only ($N = 564$); all other measures use the full sample ($N = 2{,}817$).

Table III
Controlled Regressions: Incremental R^2 of Sentiment beyond Controls (All 24 Measures)

Method	**Text Unit**	$\hat{\beta}_{sent}$	t**-stat**	p**-value**	R^2 **(M1)**	R^2 **(M3)**	ΔR^2	N
Fine-tuned DistilBERT (OOS only)								
Finetuned	Presentation	$+0.00561^{*}$	2.10	0.0354	0.1096	0.1179	0.00829	554
Finetuned	Full Call	$+0.00559^{*}$	2.10	0.0359	0.1096	0.1178	0.00824	554
Finetuned	Mgmt QA	$+0.00208$	0.89	0.3718	0.1096	0.1107	0.00115	554
Loughran-McDonald with negation								
LM_neg	Full Call	$+0.00501^{***}$	4.30	< 0.001	0.0771	0.0835	0.00638	$2,773$
LM_neg	Presentation	$+0.00428^{***}$	3.73	< 0.001	0.0771	0.0818	0.00474	$2,773$
LM_neg	Mgmt QA	$+0.00394^{***}$	3.44	0.0006	0.0771	0.0811	0.00402	$2,773$
Loughran-McDonald (plain)								
LM	Full Call	$+0.00486^{***}$	4.17	< 0.001	0.0771	0.0831	0.00599	$2,773$
LM	Presentation	$+0.00419^{***}$	3.65	< 0.001	0.0771	0.0816	0.00454	$2,773$
LM	Mgmt QA	$+0.00378^{***}$	3.34	0.0009	0.0771	0.0808	0.00370	$2,773$
FinBERT								
FinBERT	Presentation	$+0.00461^{***}$	4.18	< 0.001	0.0771	0.0826	0.00552	$2,773$
FinBERT	Full Call	$+0.00445^{***}$	4.17	< 0.001	0.0771	0.0822	0.00512	$2,773$
FinBERT	Mgmt QA	$+0.00203$	1.85	0.0638	0.0771	0.0782	0.00109	$2,773$
Harvard General Inquirer								
HV	Mgmt QA	$+0.00259^{*}$	2.30	0.0212	0.0771	0.0789	0.00178	$2,773$
HV	Full Call	$+0.00110$	0.98	0.3261	0.0771	0.0774	0.00033	$2,773$
HV	Presentation	$+0.00030$	0.27	0.7892	0.0771	0.0771	0.00002	$2,773$
Harvard General Inquirer with negation								
HV_neg	Mgmt QA	$+0.00247^{*}$	2.20	0.0276	0.0771	0.0787	0.00163	$2,773$
HV_neg	Full Call	$+0.00106$	0.95	0.3421	0.0771	0.0774	0.00030	$2,773$
HV_neg	Presentation	$+0.00024$	0.21	0.8299	0.0771	0.0771	0.00002	$2,773$
Custom Word2Vec dictionary								
Custom	Presentation	$+0.00244^{*}$	2.15	0.0314	0.0771	0.0787	0.00157	$2,773$
Custom	Full Call	$+0.00226^{*}$	2.03	0.0424	0.0771	0.0784	0.00134	$2,773$
Custom	Mgmt QA	$+0.00114$	1.06	0.2884	0.0771	0.0774	0.00035	$2,773$
Distilled DistilBERT (Knowledge Distillation)								
Distilled	Full Call	$+0.00111$	0.99	0.3216	0.0771	0.0774	0.00033	$2,773$
Distilled	Presentation	$+0.00093$	0.82	0.4144	0.0771	0.0773	0.00023	$2,773$
Distilled	Mgmt QA	$+0.00084$	0.74	0.4581	0.0771	0.0773	0.00019	$2,773$

Dependent variable: CAR-11-Carhart. M1: controls only (surp, SurpDec, NUMEST, hvol). M3: controls + sentiment. Sentiment standardised (z-score). HC3 robust standard errors.
$\Delta R^2 = R^2(\text{M3}) - R^2(\text{M1})$, computed on the same sample for both models.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
The finetuned model uses the OOS test set ($N = 554$); controls baseline $R^2 = 0.1096$ reflects this subsample.

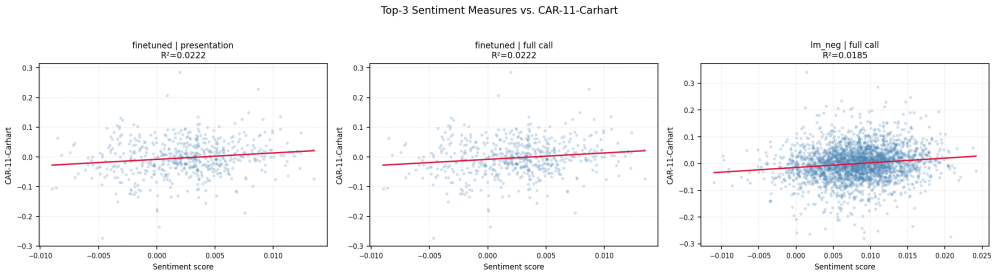

Figure 1. HOLAHOLA

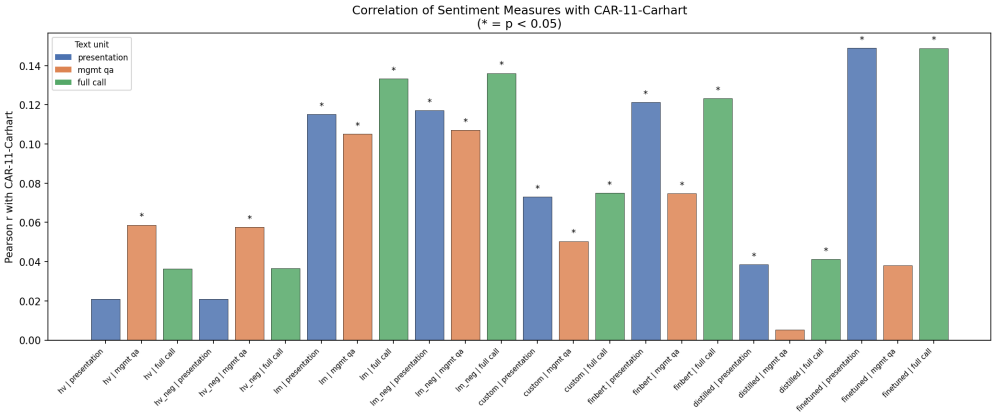

Figure 2. Biggest 3 Sentiment Measures against factor returns

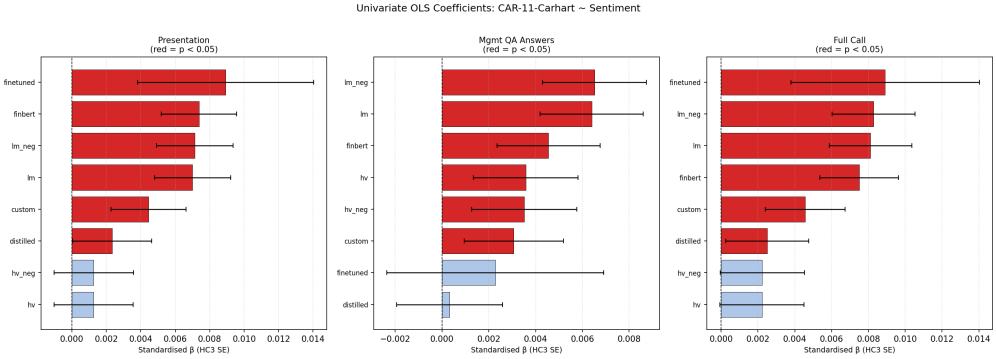

Figure 3. Correlation of Sentiment Features against factor returns

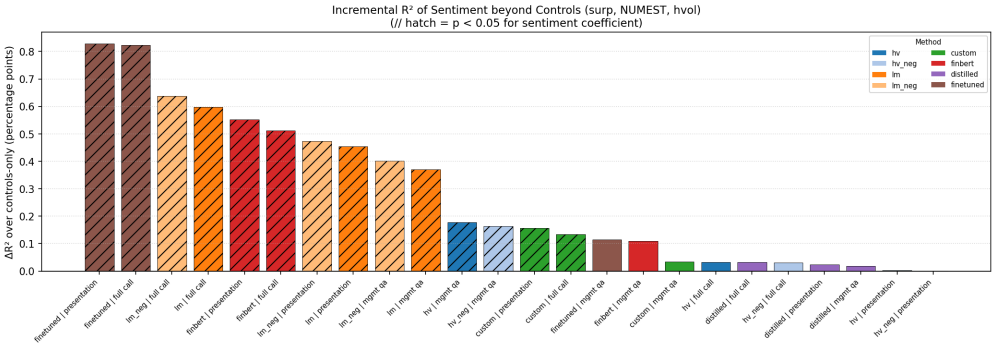

Figure 4. Coefficients of the the univariate regression model with control variables

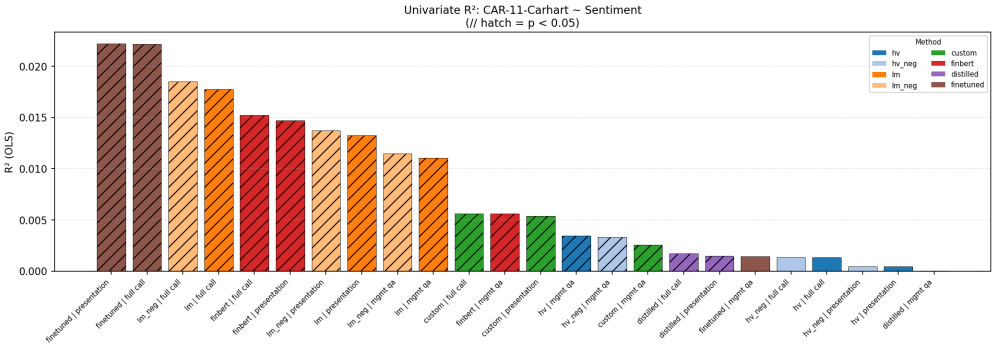

Figure 5. Change in R^2 after adding the control variables

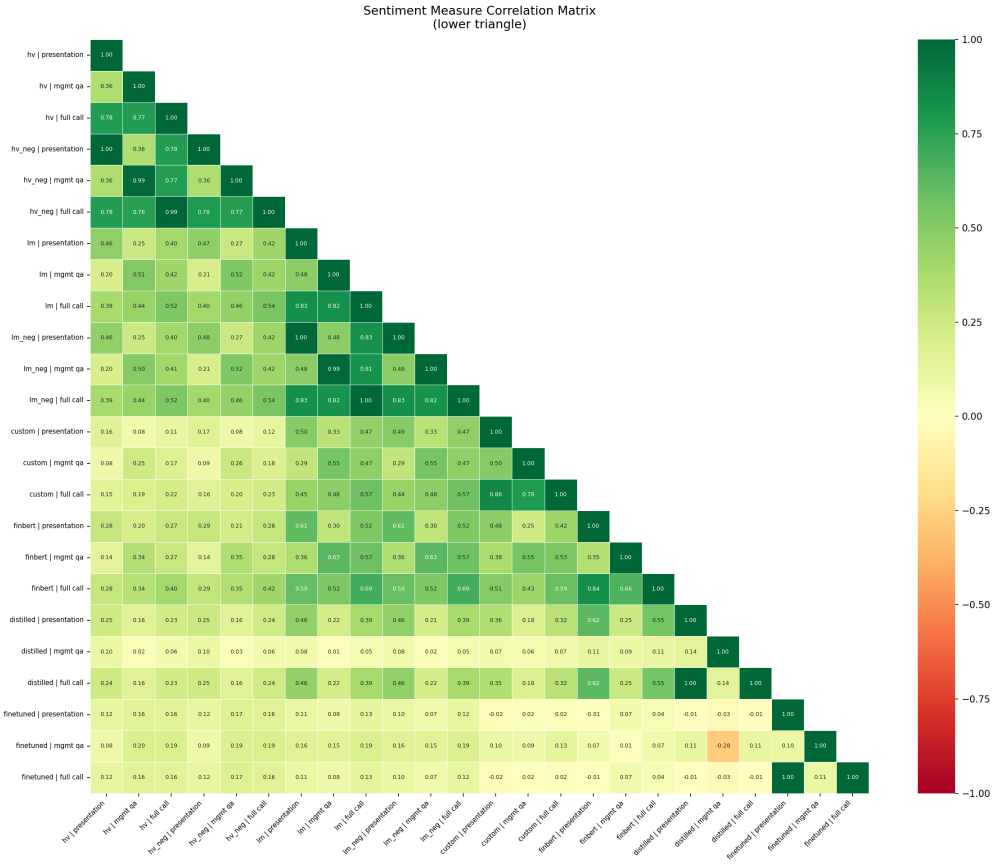

Figure 6. Univarate R^2 of single feature regression

Figure 7. Correlation of sentiment Features against control variables